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Measurement of Methane Emission from Gathering and Processing Facility: A Case Study in Typical Oil Field of China

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Abstract

Climate change has been ranked as one of the most severe challenges in the world. Greenhouse gas (GHG) emission was assumed to be the root cause. Carbon dioxide and methane are the most important GHG, while methane has a shorter atmospheric lifetime and higher global warming potential. Methane abatement was proposed as an indispensable measure to limit near-term global warming. Oil and gas (OG) production system are key methane emissions sources, with gathering and processing facility making high contribution. Measurement of methane emission from gathering and processing facility in an OG field was performed in China. Different components types, such as pressure gauge, valve, flange, connections, open-end pipeline and compressor system, were measured in 3 gathering facilities and 18 processing plants with a portable flame ionization detector (FID) and full range sampler. The component and facility based methane emission characteristics were confirmed. The results suggested 526 leak sources were identified from all sampled process components or sealing points. The leak frequency was founded to be 0.40% as a whole while it was 0.08% ~ 0.50% for gathering facility and 0.18% ~ 1.92% for processing plant, with gauge and valve category making up the highest portion. The open-end pipeline was more prone to suffer from leak problem than other type components. The flange category has a lowest population averaged emission rate of 0.16 L/min while compressor-related source has the highest mean value of 1.26 L/min. The population averaged emission rate was 0.72 L/min for gauge category and 0.22 L/min for valve category. In terms of facility-level emission rate, the population averaged value was determined to be 2.69 L/min per compressor system. The annual methane emission amount was estimated to be 45170.84 cubic meters (32.26 tons) for all sampled gathering and processing facilities. Different source category make disproportional contribution to total annual emission. The gauge category and compressor category predominates over other sources, contributing to more than 45.76% and 21.96% total emission respectively. The majority of emitted methane was attributed to a small portion of sources with emission rate higher than 1 L/min

Keywords: Methane, Emission, Oil and Gas Production system, Gathering and Processing Facility

INTRODUCTION

Climate change has emerged as one of the most severe challenges in the world. Greenhouse gas (GHG) emission was assumed to be the root cause[1]. According to recent International Energy Agency (IEA) Global methane tracker report[2], methane is responsible for around 30% rise in global temperature since the Industrial Revolution, and has been ranked as the most important GHG second to carbon dioxide. Methane, a potent GHG, has shorter atmospheric lifetime and higher global warming potential than carbon dioxide. The sustained and rapid reduction in methane emission has been regarded as one of the most important measures to limiting near-term global warming[2,3]. Therefore, methane abatement represents an indispensable portion of climate control goal of the Paris Agreement.

Methane emission was mainly caused by human activities[3-5]. According to IEA's most up-to-date report, energy system accounts for over 35% human-caused methane emissions, with oil and gas (OG) production processes emitting almost 77 Mt methane globally[2]. Methane emission from OG industry takes place along the production, gathering and processing, transmission and storage, and distribution processes in the form of fugitive emissions and vented emissions[6,7]. Unintentional leaks from faulty components or sealing points, tank-related emissions, compressor-related release, venting from pneumatic actuated device, abnormal condition-related emission, and flare were major causes of upstream methane emissions[8-19].

A few works have been undertaken on characterization of methane emission from upstream facilities of OG system, especially in United States and Canada[7-15]. The production sites or well pads have been surveyed in different OG basins. The site specific methane emission rates ranged from 0.4 to 10.5 kg/h in a recent study[15]. The component leaks at production sites in the Appalachian region were sampled, and the average methane emission rate was estimated to be 1.23 ± 0.44 g/well for component leaks[8]. Apart from production wells, hundreds of gathering facilities and processing plants in United States natural gas system were measured and the methane emission rates ranged from 0.7 to 700 kg/h for gathering facilities and 3 to 600 kg/h for processing plants[11].

The OG system emits about millions tons methane in China[2], mainly through leak and venting process. Steady increase in OG production in China calls for comprehensive understanding on methane emission. The gathering and processing facilities were primary methane emission sources. In view of the relative sparse information on methane emission from OG sources in China, especially gathering and processing operation related sources, methane emission characterization needs to be strengthened. In this work, methane emission from upstream gathering and processing facilities in an OG field was measured in China. The gathering and processing operation based emission characteristics were estimated and compared. Methane emission rate of different component category were built for the first time in China. Annual methane emission were determined for the whole sampled facilities.

METHOD

On-site measurements of methane emission from different upstream gathering and processing facilities were performed at 3 gathering facilities and 18 processing plants in an OG basin in northwest region of China. With assistance of local operators, the fugitive methane emission study was carried out between June 2022 and September 2023, using a portable flame ionization detector (FID) and a full range sampler (FRS). The detection limit of FID and FRS was 0.1 μmol/mol and 0.001 L per minute while FID has an relative standard deviation of 2% and FRS has a value of 5% respectively. The FID (Minhope Inc., Qingdao, Shandong) was adopted to identify potential leaking points and venting sources, and then emission rate was recorded with the FRS (Bacharach Inc., New Kensington, PA).

The gathering and processing operation mainly refers to collection, metering, depressurizing and purification of well produced OG flow. A sets of facilities, such as compressor, pump, multi-phase separator, gauging setup, were used in gathering and processing facilities in order to obtain natural gas product and crude stream that comply with pipeline transportation requirement. Hundreds thousands of pressure gauges, temperature gauges, valves, flanges and connectors were applied along the gathering and processing operation which also act as potential methane emission sources. In this work, all process components and sealing points in the sampled gathering and processing facilities (tank were excluded) were screened and the detailed information can be found in table 1.

Table 1—Information on process components sampled in this study

	Gathering facility	Processing plant
Gauge (pressure and temperature)	407	4290
Valve	9079	68869
Flange	351	27617
Connector	32	16342
Compressor	0	1820
Pump	0	1688
Open-end pipeline	0	33

The FID was deployed around the process components or sealing points, and air surrounding the potential emission sources were draw in together with emitted species through a sample line. According to suggestions in the emission standard of air pollutants for onshore oil and gas exploitation and production industry in China[20], concentration higher than leak standard (2000 μmol/mol) indicates the presence of leak points or emission sources. There are some components with concentration higher than 10 μmol/mol while lower than 2000 μmol/mol. These sources were also recorded and tagged. The subsequent on-site results suggest that sources with concentration lower than 2000 μmol/mol were far below detection limits of FRS. Because of the negligible role, sources with concentration lower than 2000 μmol/mol were not compared in this work.

In case of emission quantification, the source was enclosed using attachments that come with the instrument. Then, emission rate was measured by drawing air mixture (air+emitted species) from the enclosure at a high flow rate (250 L/min) to capture all the gas emitted by the leaky component. By accurately measuring the flow rate of the sampled stream and the background corrected concentration within the stream, the emission rate is calculated. The instrument was calibrated using pure methane and a mixture containing 2.5% methane. The flow meters report flows at a sampling frequency of 10 Hz. In all cases, composition characteristics were constructed with canister sampling and a GC-FID/MS analysis system to derive the methane emission rate.

RESULTS AND DISCUSSION

Leak frequency

Around 130 528 process components and sealing points were measured in 3 gathering facilities and 18 processing plants. The FID survey results suggested all facilities were suffering from methane emission problem and 526 leak sources were identified. The gathering and processing facility make disproportional contribution to emission sources. About 39 sites stem from gathering facility while processing plants contributes the rest 487 leaking points.

Leak frequency of facility was given in figure 1. The leak frequency was estimated to be 0.40% as a whole while significant difference were observed among gathering facility (1# ~ 3#) and processing plants (4# ~ 21#). The gathering facility has a leak frequency of 0.08% ~ 0.50% while it appears that processing plants have higher leak frequency, and almost 0.17% ~ 1.92% of seals and process components were labeled as emission sources.

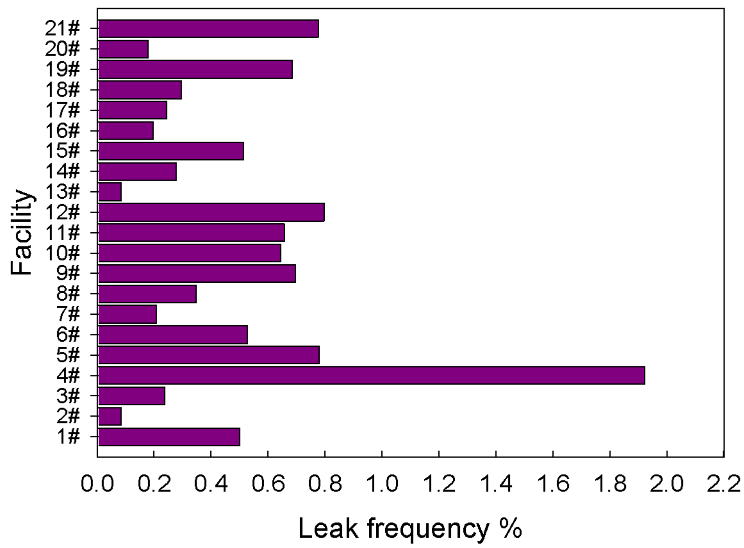

Figure 1—Leak frequency of gathering and processing facilities

Composition of emission sources was confirmed. Process component makes distinct contribution to emission sources. The gauge (pressure gauge and temperature gauge) and valve category accounts for 32.89% and 34.60% of emission sources as a whole. Compressor-related components are responsible for 13.31% of emission sources. Flange and connector also play an important role in affecting methane leak process. Around 9.13% and 6.27% of leak points were attributed to flange and connector respectively.

The possibility of process component that tends to cause methane leak were determined as given in figure 2. The open-end line has much higher possibility to leak, and over 30% percent of open-end line population suffer from leaking problem. The gauge category and compressor system were also prone to generate methane emission though component leak process, and almost 3.64% of gauge category and 3.85% of compressor-related component appear to suffer from leaking problem. Comparatively, the flange, valve, connector and pump category are less prone to cause methane leak.

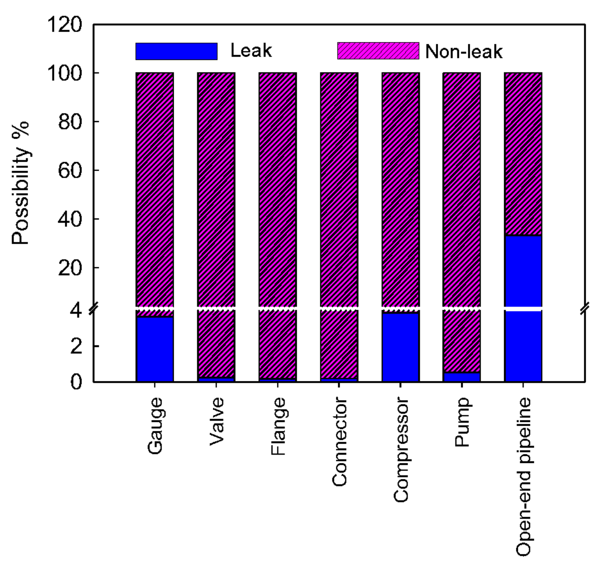

Figure 2—Tendency of process components to leak

Methane emission rate

Because of the detection limit of FRS, only sources with emission rate higher than 0.001 L per minute can be quantified by the FRS measurement. Among the identified 526 sources, a total of 153 leak points were quantified by FRS measurement and the methane emission rate was consequently estimated. The FRS failed to function in nearly 373 sources. The obtained emission rate for specific source category, such as gauge (pressure gauge and temperature gauge), flange, valve, compressor-related components and open-end pipeline were compared in table 2. A discrete distribution characteristics was observed among individual emission sources. In some cases, only several milliliter methane was emitted per minute while thousands of milliliters methane were vented for specific emission sources per minute.

Table 2—Methane emission rate for different process component

Process component	Rate range L per minute	Population averaged rate L per minute	Total emission rate L per minute
Gauge	2.79×10^{-3}~13.11	0.72	39.33
Valve	1.89×10^{-3}~0.99	0.16	3.10
Flange	1.62×10^{-3}~2.36	0.22	11.65
Compressor-related	2.01×10^{-3}~8.12	1.26	18.88
Open-end pipeline	2.32×10^{-3}~7.92	1.18	12.99

Further analysis suggests that compressor-related leak components have an averaged methane emission rate of 1.26 L/min, which was relative higher than other source category. The methane emission rate ranges from 2.32×10^{-3} L/min to 7.92 L/min for open-end pipeline, with an averaged value of 1.18 L/min. In terms of gauge and valve category, the averaged emission rate was 0.72 L/min and 0.22 L/min respectively. The flange category has a much lower averaged value, which was estimated to be 0.16 L/min. The average emission rate for pneumatic controllers and equipment leaks were compared in previous report[8], which has a value of 3.8 ~ 5.6 L per minute and 1.1 ~ 2.3 L per minute. The component leak rate in China was comparable with that of USA, while much lower than pneumatic controllers.

By linear sum of compressor-related leaking source, compressor-level methane emission rate was estimated. In summary, the specific emission rate for compressor ranged from 0.15 L/min to 8.16 L/min for the whole group (8 compressor), with an arithmetic mean value of 2.69 L/min. Further analysis suggest that a total of 4 sampled compressors have a rate higher than 1 L/min while the rest 3 compressors have an emission rate lower than 1 L/min.

Composition characteristics of sources emission rate were discussed as shown in figure 3. The emission sources were divided into three groups according to rate value, low emission group (LEG, emission rate lower than 0.1 L/min), medium emission group (MEG, emission rate between 0.1 L/min and 1 L/min) and high emission group (HEG, emission rate higher than 1 L/min). As a whole, most of the processing facility related leak sources was recognized as LEG and MEG, contributing 91.5% of total emission sources by population. The same phenomena was observed in specific sources category, with HEG making up increased contribution to compressor and open-end pipeline related sources.

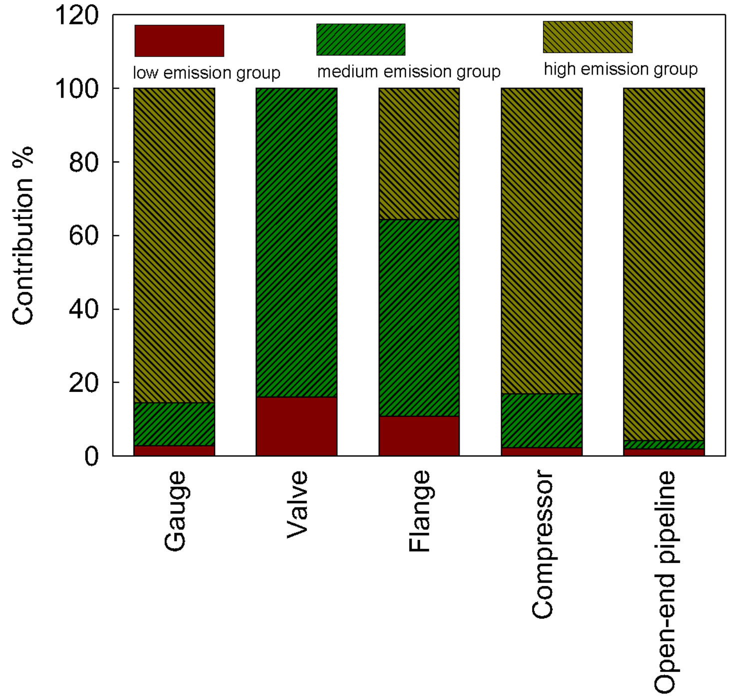

Figure 3—Contribution to total emission rate for different source category

The cumulative emission rate from all sources amounts to 39.33 L/min (gauge category), 3.10 L/min (flange category), 11.65 L/min (valve category), 18.88 L/min (compressor-related sources) and 12.99 L/min (open-end pipeline category). A disproportional distribution characteristics was observed among emission sources, with a small fraction of components accounting for the majority of methane emission. For example, the HEG sources (6 out of 55) contribute nearly 85.60% of total emission for gauge category, while it only makes up 2.84% of gauge population. Similarly, the HEG sources contribute 35.68%, 83.19%, and 95.76% of total emission for valve category (2 out of 54), compressor-related sources (3 out of 15) and open-end pipeline category (3 out of 11). This finding coincide with results reported by previous work[11,15].

Annual methane emission

Considering the relevant constant operation conditions, the annual methane emission amount was projected under a robust assumption that sources emission rate over longer time range can be represented by the snapshot-in-time sample period. In summary, the sampled gathering facility and processing plant yield an annual methane emission of 45170.84 cubic meters (32.26 tons), and the sample population averaged annual value was 2150.99 cubic meters (1.54 tons) per facility and 295.23 cubic meters (0.21 tons) per emission source.

The contribution from specific source groups differs significantly. Detailed information suggests that gauge category and compressor category act as primary contributors, accounting for 45.76% and 21.96% of annual emission. The valve and open-end pipeline category also make important contribution, causing 13.55% and 15.12% annual emission. In contrary, flange category makes negligible contribution. Further analysis suggest that 14 specific sources contributes nearly 77.06% of total methane, and all of the sources were HEG in nature. The results suggest that targeted control of HEG will be key and effective measures to reducing methane emission, especially component leaks from gathering and processing operation.

The annual emission intensity (ton/a) for different source category were build and compared in table 3. The intensity for specific sources differs in orders of magnitude. Regardless of sources category, the HEG has highest value while MEG sources have an intensity ten times higher than LEG. The sample population averaged value was 2.11 tons, 0.78 tons, 1.97 tons and 1.56 tons for gauge, valve, compressor-related components and open-end pipeline respectively.

Table 3—Annual methane emission intensity for different process component

Process component	low emission group (tons)	medium emission group (tons)	high emission group (tons)
Gauge	1.19×10^{-2}	0.12	2.11
Valve	1.69×10^{-2}	0.14	-
Flange	1.40×10^{-2}	0.13	0.78
Compressor-related	1.98×10^{-2}	0.26	1.97
Open-end pipeline	1.49×10^{-2}	0.06	1.56

Mitigation measures

Component leaks will result in substantial methane emission from oil and gas operations process, and leak detection and repair (LDAR) programme were proposed as most effective measures to drive down these losses. Based on findings in this work, targeted leak detection focused on gauge and open-end pipeline with high frequency (four times per year) were suggested with portable FID. The leak concentration can be used to screen out sources with high emission rate (always characterized by concentration higher than 10000 μmol/mol). Identification, location and subsequent mitigation of these high emission sources through prompt repair, leaky component replacement, and operation optimization will play critical role in methane emission reduction.

CONCLUSIONS

For the first time, methane emission from upstream gathering and processing facilities in typical OG field of China were measured. The emission characteristics were analyzed, and emission rate for specific sources were determined. Nearly all facilities were suffering from methane emission problem and 526 leak points or emission sources were identified. Source specific emission rate ranges from 1.62×10^{-3} L/min to 13.11 L/min, while it differs for specific sources category. The compressor-related sources have an population averaged methane emission rate higher than other category. Cumulative emission rate for specific source category amount to 3.10 ~ 39.33 L/min, and a high skew distribution was observed with small portion of high-emitting sources contributing the majority of emission. Annual mass methane emission was estimated to be 32.26 tons, with sample population averaged annual value of 1.54 tons per facility and 0.21 tons per emission source. Gauge category and compressor category act as primary contributor to annual emission and most of the emitted methane was attributed to sources with emission rate higher than 1 L/min. Targeted leak detection with high frequency and prompt control of sources with high emission rate were proposed to efficiently drive down methane emission. The information obtained in this study improves our understanding on methane emission from upstream gathering and processing operations in China and will play important role in further estimation of methane emission from OG gathering and processing process.

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